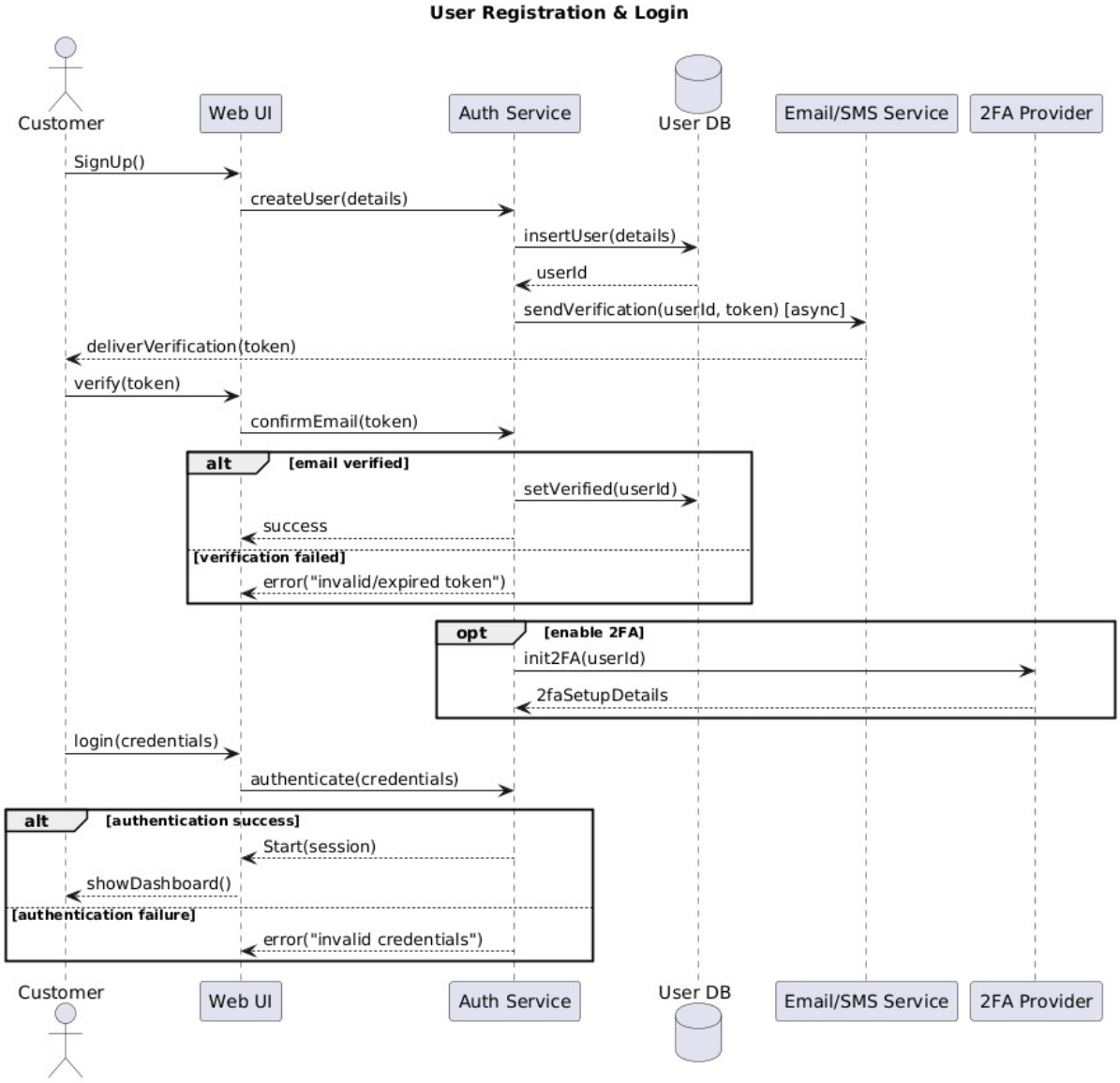
SmartBooks: An Online Bookstore System

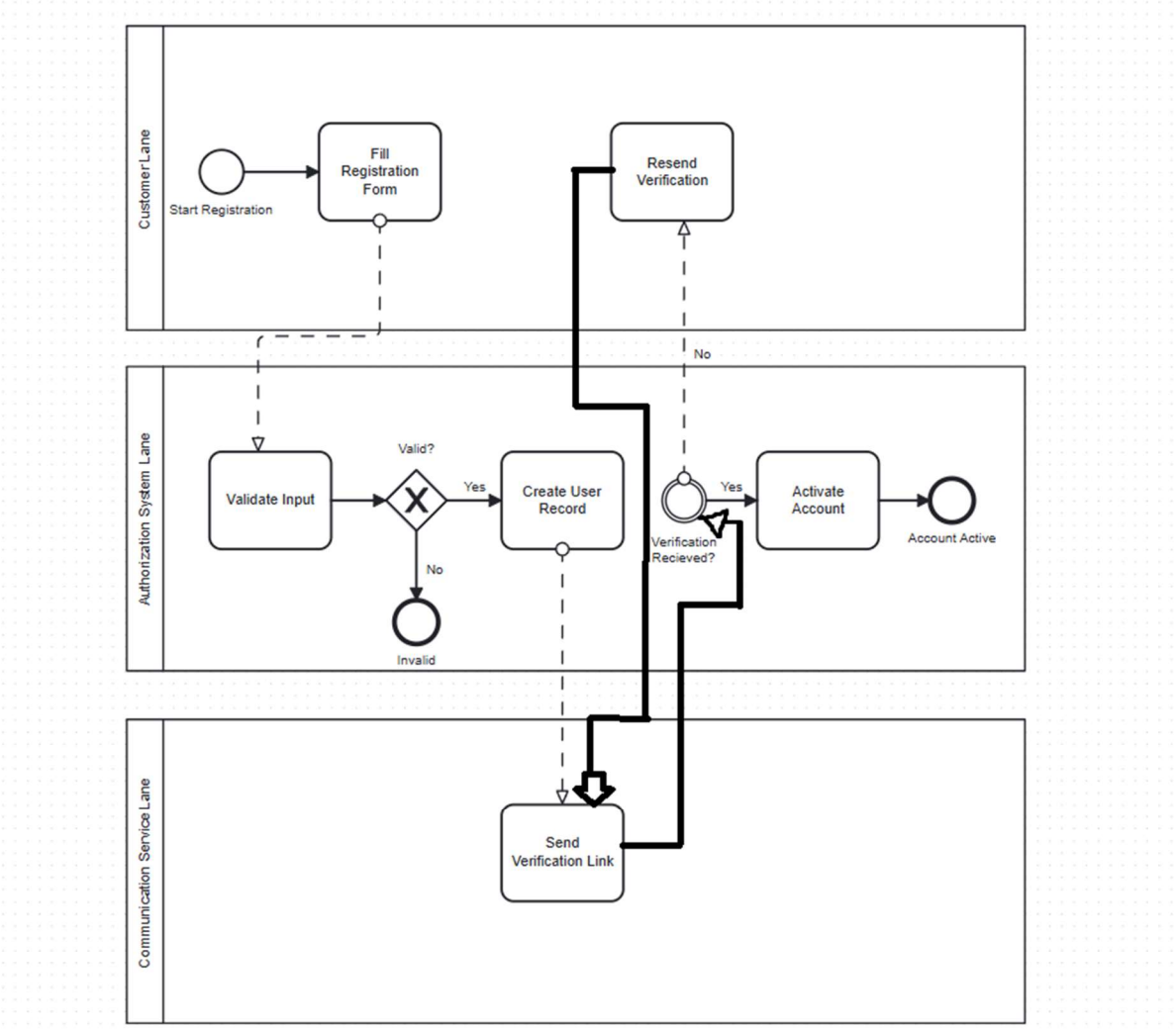
SmartBooks is an online bookstore that lets customers create accounts, search and browse the catalog, add books to a cart, place and pay for orders, and track deliveries while admins manage stock and suppliers. The system is designed for secure, high-availability operation and a responsive UI that scales for peak traffic — the provided UML/BPMN diagrams document the core workflows and interactions required for implementation and submission.

1) UML Sequence Diagram (Ahmed Ramy): User Registration & Login:



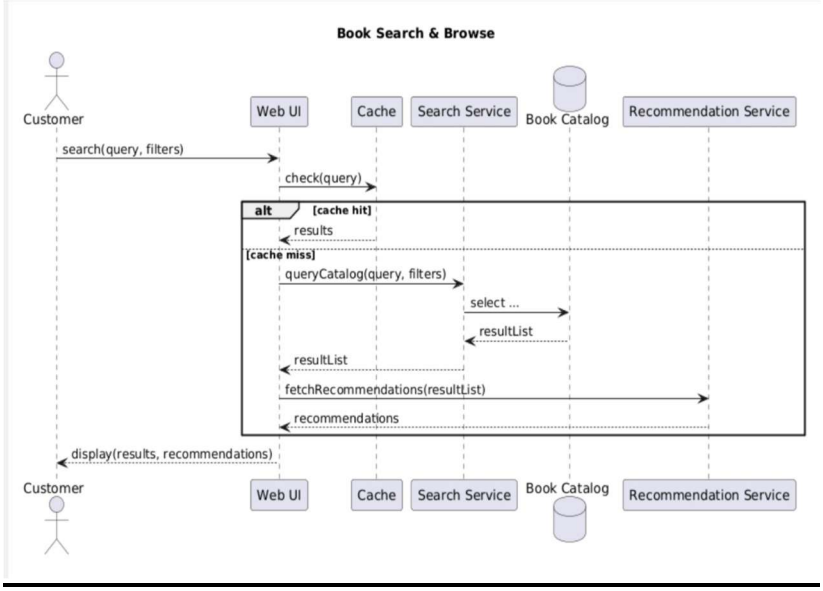
Purpose: Sequence showing user sign-up, email verification, optional 2FA, and login.

1) BPMN: User Onboarding (Registration → Verification → Activation):



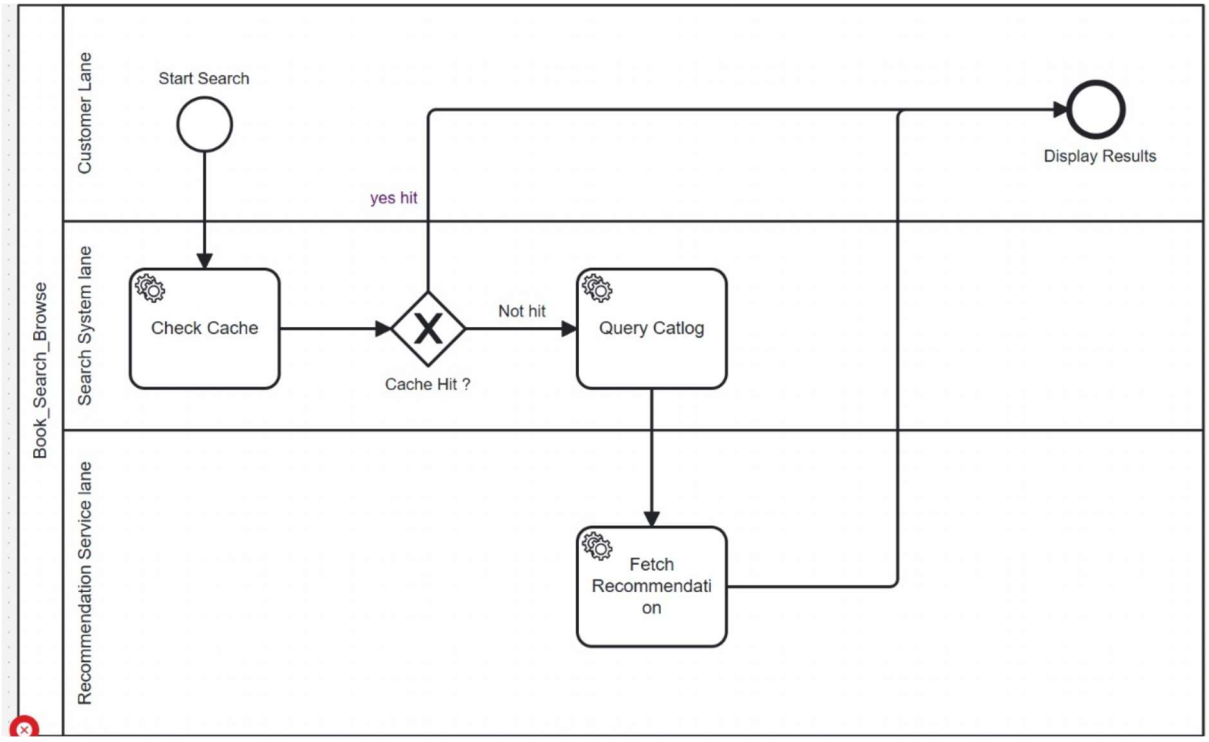
Purpose: business-level flow for new user onboarding.

2) **UML Sequence Diagram (Ahmed Amr):** Book Search & Browse:



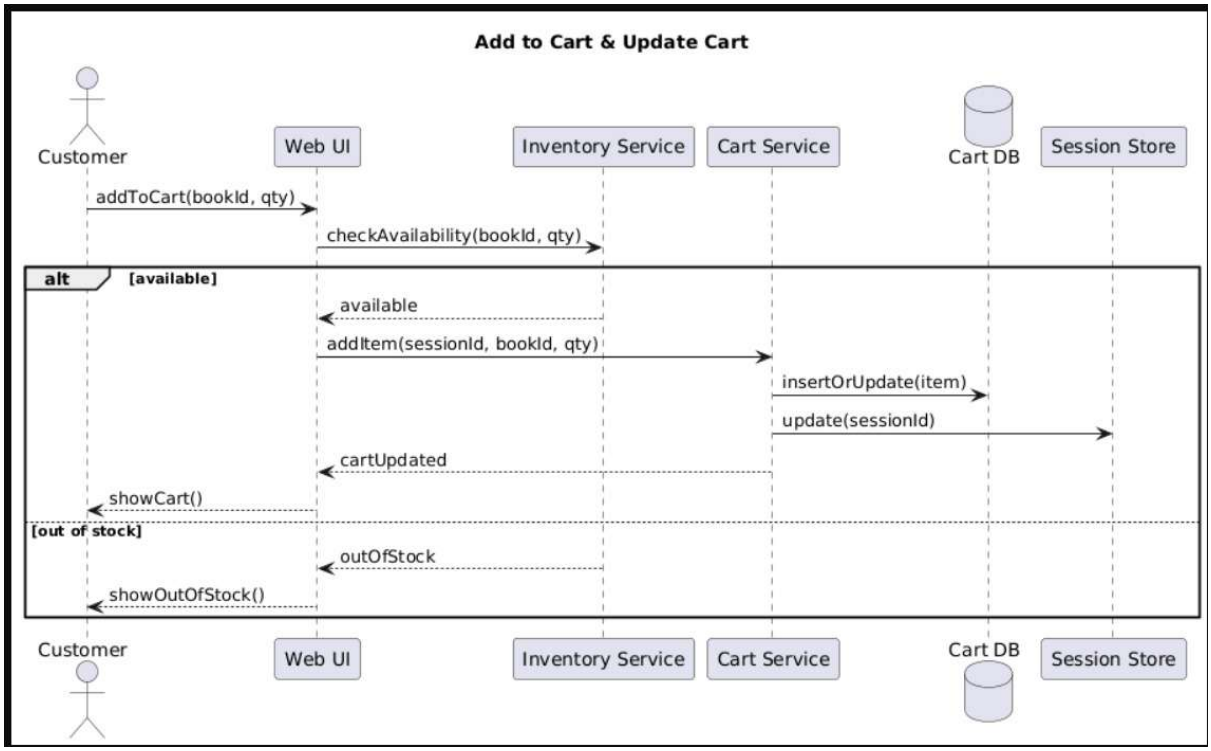
Purpose: Sequence for user registration flow**.**

2) BPMN: Book Search & Browse (Cache → Query → Recommendations):



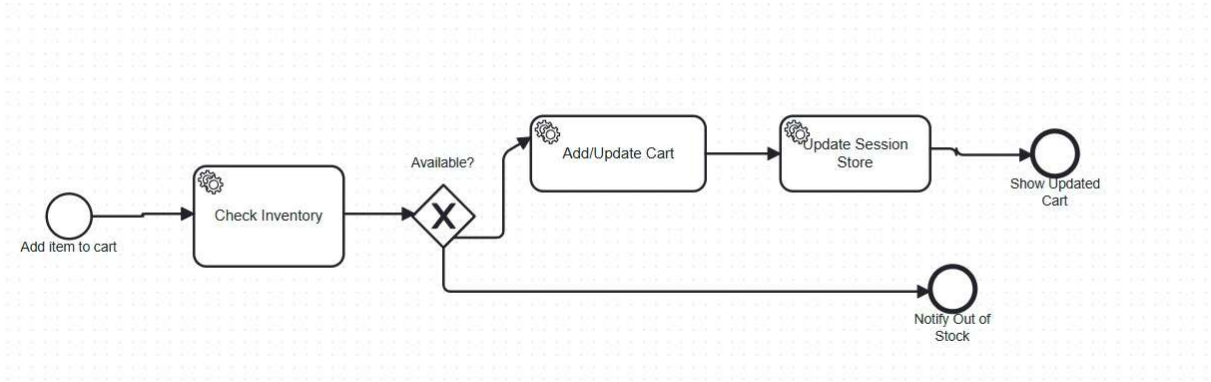
Purpose: Business flow for user onboarding.

3) **UML Sequence Diagram (Ramez Mokbil)**: Add to Cart & Update Cart:



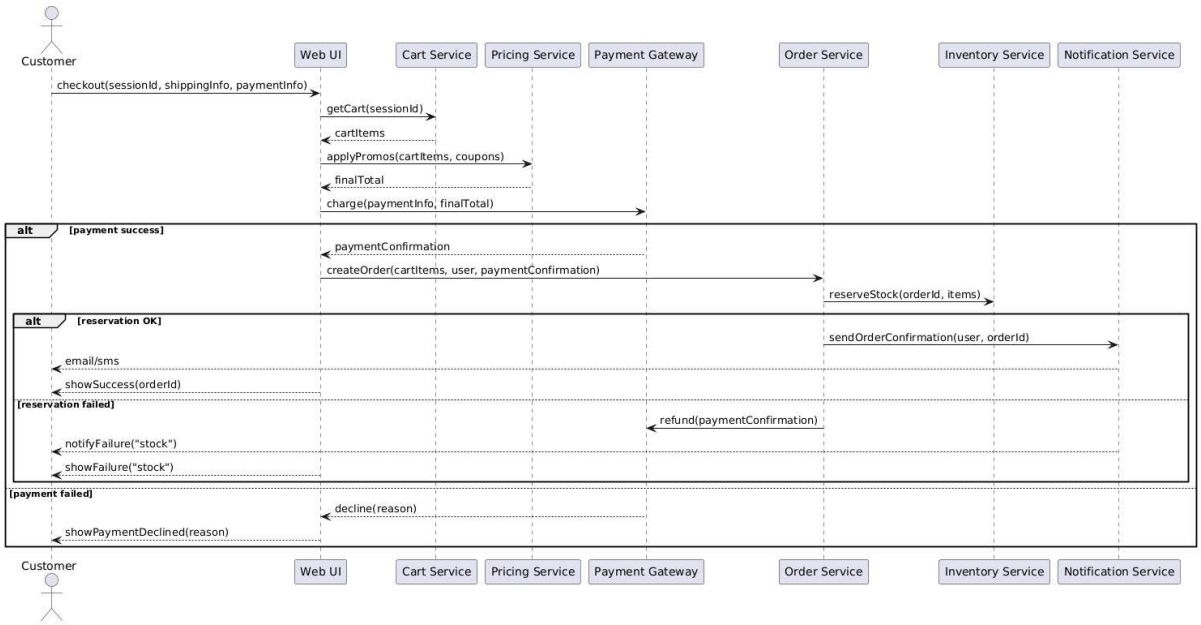
Purpose: Capture adding/removing items and quantity update (cart management).

3) **BPMN**: Inventory Replenishment & Supplier Coordination:



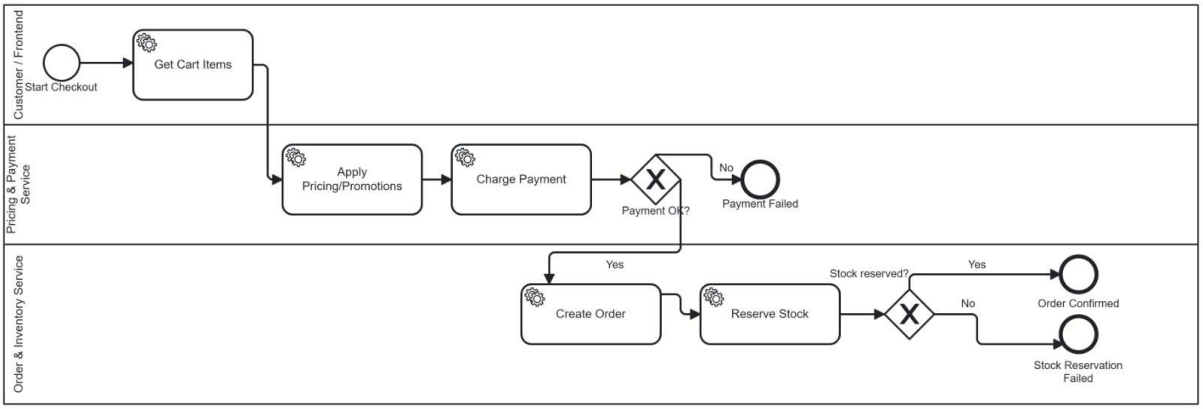
Purpose: Model admin-side inventory checks (maps to admin/manage stock requirement).

4) **UML Sequence Diagram (Seif Tarek)** : Checkout & Payment:



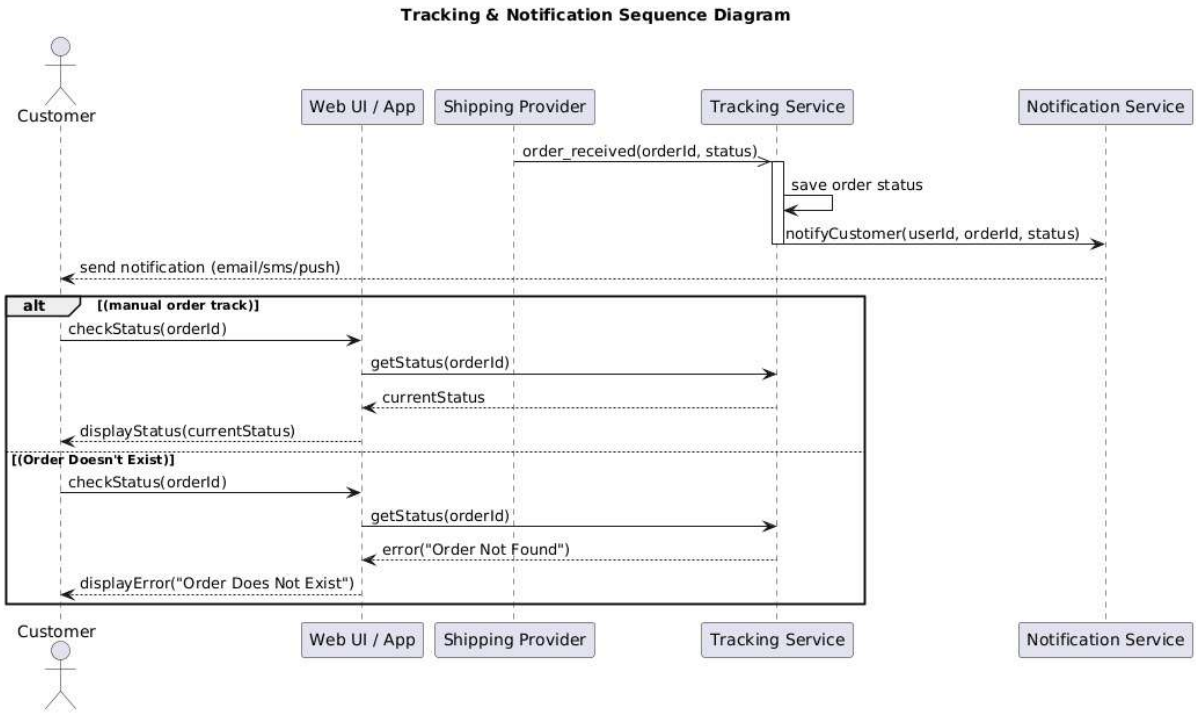
Purpose: Full checkout flow —shipping selection, payment, order creation.

4) **BPMN**: Checkout & Order Processing Workflow:



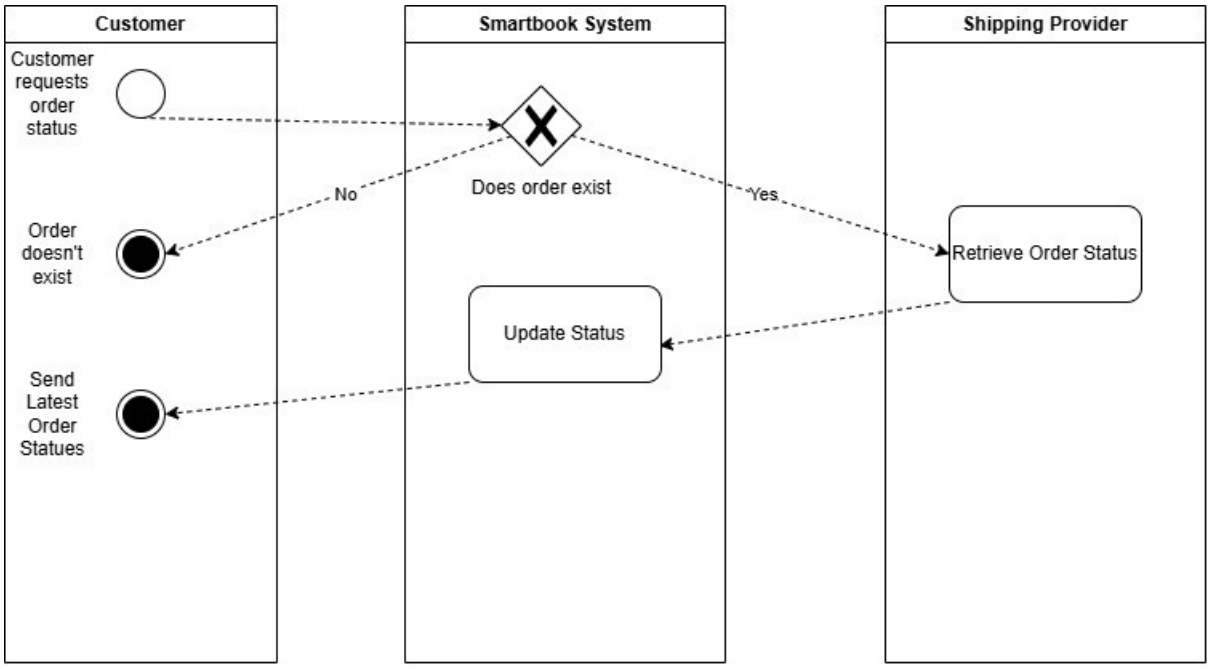
Purpose: Modelling the end-to-end checkout flow, including pricing calculation, payment processing, order creation, and stock reservation.

5) **UML Sequence Diagram (Saif Saad)**: Order Tracking & Notifications:



Purpose: show how order status updates flow from shipping to the customer.

5) BPMN: Order Status Retrieval & Update Workflow:



Purpose: This BPMN models how the SmartBooks system handles a customer's request to check an order's status.